

Experiment 3: Insertion Sort – Real-Time Stock Price Feed

Aim

To implement the **Insertion Sort** algorithm in Python to insert new stock prices into an already sorted list.

Problem Statement

A trading dashboard receives stock prices one by one and must maintain them in sorted order for quick analysis. Use **Insertion Sort** to insert each new stock price into its correct position.

Algorithm

Step 1: Start.

Step 2: Read the number of stock prices.

Step 3: Read the stock prices into an array.

Step 4: Apply Insertion Sort.

Step 5: Compare the current price with previous prices.

Step 6: Shift larger elements one position to the right.

Step 7: Insert the current price in its correct position.

Step 8: Display the sorted stock prices.

Step 9: Stop.

Source Code (Python)

```
main.py
1 # Real-Time Stock Price Feed using Insertion Sort
2
3 n = int(input("Enter number of stock prices: "))
4 prices = []
5
6 print("Enter stock prices:")
7 for i in range(n):
8     prices.append(float(input()))
9
10 for i in range(1, n):
11     key = prices[i]
12     j = i - 1
13
14     while j >= 0 and prices[j] > key:
15         prices[j + 1] = prices[j]
16         j -= 1
17
18     prices[j + 1] = key
19
20 print("Sorted Stock Prices:", prices)
21 print("Minimum Price:", prices[0])
22 print("Maximum Price:", prices[-1])
```

Output

Output
Enter number of stock prices: 5 Enter stock prices: 102.5 98.3 105.1 100.0 97.8 Sorted Stock Prices: [97.8, 98.3, 100.0, 102.5, 105.1] Minimum Price: 97.8 Maximum Price: 105.1 === Code Execution Successful ===

Complexity Analysis

Aspect	Complexity
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Time Complexity	$O(n^2)$
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Space Complexity	$O(1)$
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Result

The Insertion Sort algorithm was successfully implemented to maintain stock prices in sorted order. The program correctly displayed the sorted prices along with the minimum and maximum stock prices.